Year 13 Further Maths - Lines and planes in space

Similar questions - Intersections and angles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The line l has equation $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ and the plane Π has equation $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 7$.

(a) Find the position vector of the point where l meets Π. **[3]**

(b) Find the acute angle between l and Π. **[3]**

(c) A is the point on l where $\lambda = 1$. The line l_2 passes through A, is perpendicular to l and is parallel to Π. Find a vector equation of l_2. **[4]**

2 The lines l_1 and l_2 intersect and have equations $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ a \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ -3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 5 \\ 2 \end{pmatrix}$ where a is a constant. The plane Π has equation $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1$. l_1 meets Π at Q and l_2 meets Π at R.

(a) Find a. **[3]**

(b) Verify that the coordinates of R are $(9, -8, 0)$. **[2]**

(c) Determine the exact length of QR. **[5]**

3

(a) Determine the acute angle between the line $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} \sqrt{6} \\ 2 \\ \sqrt{6} \end{pmatrix}$ and the y-axis. **[4]**

(b) The plane Π_1 has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$. Find the acute angle between Π_1 and the plane with equation $\mathbf{r} \cdot \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = 5$. **[4]**

(c) The point A has coordinates $(7, -1, 9)$. Find the coordinates of the foot F of the perpendicular from A to Π_1. **[4]**

Question 1

(a) $3(2+\lambda)-(-1+2\lambda)+2(3-2\lambda)=7$ gives $13-3\lambda=7$, $\lambda=2$. Point: $4\mathbf{i}+3\mathbf{j}-\mathbf{k}$

(b) $\sin\phi=\dfrac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}=\dfrac{3}{3\sqrt{14}}$, so $\phi=15.5^\circ$

(c) $A=(3,1,1)$. The direction is perpendicular to $\mathbf{d}$ and $\mathbf{n}$: $\mathbf{d}\times\mathbf{n}=(2,-8,-7)$. $l_2:\mathbf{r}=\begin{pmatrix}3\\1\\1\end{pmatrix}+\mu\begin{pmatrix}2\\-8\\-7\end{pmatrix}$

Question 2

(a) Equating x: $1+2\lambda=7-2\mu$; y: $\lambda=-3+5\mu$. So $1-6+10\mu=7-2\mu$... giving $\mu=1$, $\lambda=2$. z: $a-2=2+2=4$, so $a=6$ (intersection $(5,2,4)$)

(b) $\mu=-1$ gives $(7+2,-3-5,2-2)=(9,-8,0)$, and $9-8+0=1$, so it lies on Π and on l_2

(c) l_1 meets Π: $(1+2\lambda)+\lambda+(6-\lambda)=1$, so $7+2\lambda=1$, $\lambda=-3$, $Q=(-5,-3,9)$. $\overrightarrow{QR}=(14,-5,-9)$, $|QR|=\sqrt{196+25+81}=\sqrt{302}$

Question 3

(a) $\cos\alpha=\dfrac{|\mathbf{d}\cdot\mathbf{j}|}{|\mathbf{d}|}=\dfrac{2}{\sqrt{16}}=\dfrac{1}{2}$, so $\alpha=60^\circ$

(b) Normal to Π_1: $(1,2,0)\times(0,1,3)=(6,-3,1)$. $\cos\theta=\dfrac{|12+0+1|}{\sqrt{46}\sqrt{5}}=\dfrac{13}{\sqrt{230}}$, so $\theta=31.0^\circ$

(c) Π_1: $6x-3y+z=8$. $F=\mathbf{a}-t\mathbf{n}$ with $6(7-6t)-3(-1+3t)+(9-t)=8$, so $54-46t=8$, $t=1$. $F=(1,2,8)$